

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 10: Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Theorem 1: tangent and radius

1. Tangents are drawn to a circle at the two ends of a diameter. How are the two tangents related?

CBSE 2025, 1 mark

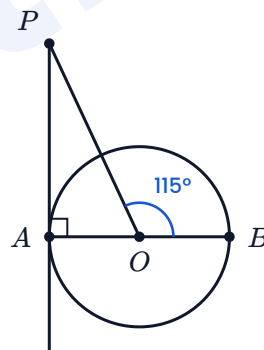
SOLUTION

Each tangent is perpendicular to the radius at its point of contact, so both tangents are perpendicular to the same diameter.

Two lines perpendicular to the same line are parallel.

Answer: the tangents are parallel

2. AOB is a diameter of a circle with centre O , and PA is the tangent at A . If $\angle POB = 115^\circ$, find $\angle APO$. CBSE 2025, 1 mark



SOLUTION

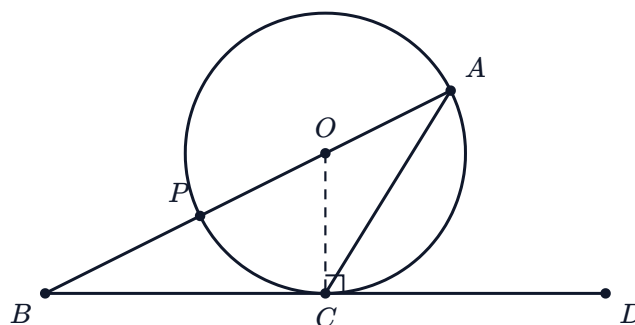
$\angle AOP = 180^\circ - 115^\circ = 65^\circ$ (angles on the straight line AOB)

$\angle OAP = 90^\circ$ (the tangent is perpendicular to the radius at the point of contact)

In $\triangle OAP$: $\angle APO = 180^\circ - 90^\circ - 65^\circ$

Answer: 25°

3. O is the centre of a circle and BCD is the tangent to it at C . The line from B through O meets the circle at P and A . Prove that $\angle BAC + \angle ACD = 90^\circ$. **CBSE 2025, 2 marks**



SOLUTION

$OA = OC$ (radii), so $\angle OAC = \angle OCA \dots (1)$

$OC \perp BD$, because the radius is perpendicular to the tangent at the point of contact. So $\angle OCA + \angle ACD = 90^\circ \dots (2)$

A, O and B lie on one line, so $\angle BAC = \angle OAC$.

From (1) and (2): $\angle BAC + \angle ACD = 90^\circ$. **Proved.**

4. Assertion (A): the tangents at the two ends of a diameter of a circle are parallel. Reason (R): the diameter is the longest chord of a circle. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true.

CBSE 2024, 1 mark

SOLUTION

A is true: each tangent is perpendicular to the diameter at its end point, and two lines perpendicular to the same line are parallel.

R is true, but it is not the reason why the tangents are parallel.

Answer: (b)

2 Theorem 2: tangents from an external point

5. A point P is 26 m from the centre of a circle. Each tangent from P to the circle is 10 m long. Find the radius of the circle. **CBSE 2025, 2 marks**

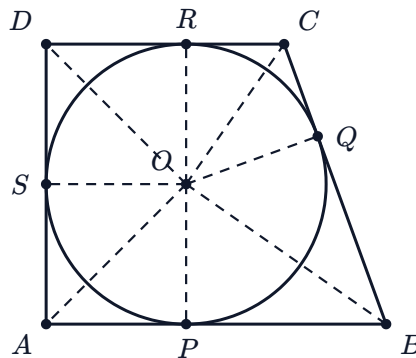
SOLUTION

Let O be the centre and T a point of contact. Then $\angle OTP = 90^\circ$.

$$OT^2 = OP^2 - PT^2 = 26^2 - 10^2 = 676 - 100 = 576$$

Answer: radius = 24 m

6. A quadrilateral is drawn around a circle so that all four sides touch it. Prove that each pair of opposite sides subtends supplementary angles at the centre. **CBSE 2025, 2 marks**



SOLUTION

Let $ABCD$ touch the circle, centre O , at P, Q, R and S on AB, BC, CD and DA .

In $\triangle OAP$ and $\triangle OAS$: $OP = OS$ (radii), $AP = AS$ (tangents from A) and OA is common. So the triangles are congruent (SSS) and $\angle AOP = \angle AOS = a$.

In the same way, $\angle BOP = \angle BOQ = b$, $\angle COQ = \angle COR = c$ and $\angle DOR = \angle DOS = d$.

The angles around O add up to 360° : $2a + 2b + 2c + 2d = 360^\circ$, so $a + b + c + d = 180^\circ$.

$\angle AOB = a + b$ and $\angle COD = c + d$, so $\angle AOB + \angle COD = 180^\circ$. Then $\angle BOC + \angle DOA = 360^\circ - 180^\circ = 180^\circ$ as well.

So opposite sides subtend supplementary angles at the centre. Proved.

7. AB is a diameter of a circle with centre O . The tangents at A and B are met by a third tangent at Q and P respectively. Prove that $\angle POQ = 90^\circ$. **CBSE 2024, 3 marks**

SOLUTION

Let PQ touch the circle at C . Join OC .

In $\triangle OAQ$ and $\triangle OCQ$: $OA = OC$ (radii), OQ is common and $QA = QC$ (tangents from Q). So $\triangle OAQ \cong \triangle OCQ$ (SSS), and $\angle AOQ = \angle COQ$.

Similarly $\triangle OBP \cong \triangle OCP$, so $\angle BOP = \angle COP$.

AOB is a straight line: $\angle AOQ + \angle COQ + \angle COP + \angle BOP = 180^\circ$, so $2\angle COQ + 2\angle COP = 180^\circ$.

Hence $\angle POQ = \angle COQ + \angle COP = 90^\circ$.

8. A circle of radius 8 cm with centre O touches the sides AD , DC , CB and BA of a quadrilateral $ABCD$ at P , Q , R and S . $AD \perp DC$, $BC = 30$ cm and $BS = 24$ cm. Find DC . **CBSE 2024, 3 marks**

SOLUTION

Tangents from B are equal: $BR = BS = 24$ cm, so $CR = BC - BR = 30 - 24 = 6$ cm.

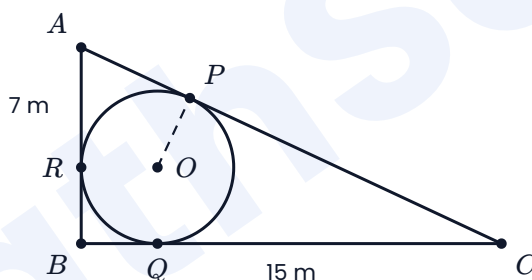
Tangents from C are equal: $CQ = CR = 6$ cm.

In quadrilateral $OPDQ$: $\angle D = 90^\circ$ (given) and $\angle OPD = \angle OQD = 90^\circ$ (radius is perpendicular to tangent), and $OP = OQ = 8$ cm. So $OPDQ$ is a square and $DQ = 8$ cm.

$$DC = DQ + QC = 8 + 6$$

Answer: $DC = 14$ cm

9. A triangular backyard ABC has a right angle at B , with $AB = 7$ m and $BC = 15$ m. A circular pit inside it touches AC , BC and AB at P , Q and R , and $AP = x$ m. (i) Write AR in terms of x . (ii) What kind of quadrilateral is $BQOR$? (iii) Write PC in terms of x and find x , OR find x and then the radius r of the pit. **CBSE 2024, 4 marks, case study**



SOLUTION

(i) Tangents from A are equal: $AR = AP = x$ m.

(ii) $\angle B = \angle OQB = \angle ORB = 90^\circ$ and $OQ = OR$ (radii), so $BQOR$ is a square.

(iii) $BR = AB - AR = 7 - x$, so $BQ = 7 - x$ and $QC = 15 - (7 - x) = 8 + x$. Tangents from C are equal, so $PC = QC = (8 + x)$ m.

$$AC = \sqrt{7^2 + 15^2} = \sqrt{274} \text{ m, and } AC = AP + PC = 2x + 8. \text{ So } x = \frac{\sqrt{274} - 8}{2} \approx \frac{16.55 - 8}{2} \approx 4.28.$$

$$\text{OR: the radius equals the side of the square, } r = BR = 7 - x = \frac{22 - \sqrt{274}}{2} \approx 2.72 \text{ m.}$$

Answer: (i) x m (ii) a square (iii) $PC = (8 + x)$ m, $x \approx 4.28$; or $r \approx 2.72$ m

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